Quick guides to the evaluation of orebodies

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ABSTRACT

Annual mining revenue can be computed by formulae relating metal prices, smelting schedules, concentrate grade and recovery to known ore grade and an assessment of core samples. Operating costs at different daily tonnages are related to orebody shape, mining method, milling process and general plant services. Capital costs are related to sizing of mining equipment, mine development, plant-site topography, climate and accessibility, plant services and personnel housing.

Introduction

A crude guide to the average 1978 cost of mine projects is shown in Figure 1, but the costs of many mine projects have differed widely from the graphed average costs of $800,000 $T^{0.6}$ for underground mine/mill projects and $400,000 $T^{0.6}$ for low-grade open-pit mine/mill projects.

High mine project costs and high operating costs are characteristic of open-pit mines with high stripping ratios, underground mines with thin orebodies, mills with complex metallurgy, and projects located in isolated regions with severe climate, mountainous topography, and lacking access to existing roads, towns and electric power. When none of these adverse conditions prevailed, the mine project cost was typically much less than the graphed average cost. Virtually all the conditions likely to cause high capital and operating costs are known from knowledge of the local topography, climate or accessibility, or can be assessed by tests on core samples, as soon as the orebody has been outlined by drilling.

A more accurate estimate of over-all project capital cost and operating cost can be made from a summation of items of cost after judging the effect of specific local conditions on each item of capital and operating cost. This paper offers guides in judging the effect of local conditions on the sizing and 1978 cost of specific items of capital cost and operating cost and operating cost of mine projects. When the specific items of capital cost and operating cost are totalled and measured against expected net revenue and taxes over the life of the mine using the net-present-value or discount-cash-flow methods, the result should show whether the mine project development is clearly feasible, doubtfully feasible or clearly uneconomic. If the development of the mine project is clearly feasible, or probably feasible, a detailed feasibility study by experienced consulting engineers should be commissioned before any financial commitments are made; but if project development appears to be uneconomic, the time and cost of a detailed feasibility study is not warranted.

The chief variable affecting all items of capital cost, revenue and operating cost is the daily tonnage of ore that is treated by the process plant. The guides to capital and operating cost offered in this paper are based on the assumptions that the process plant will be operated continuously for seven days per week, but mining operations and crushing plants will operate for only five days per week, with only necessary maintenance services on weekends. Thus, the daily ore tonnage mined and crushed will be 40% greater than the daily ore tonnage milled.



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T. Alan O'Hara was born in New Zealand and graduated with B.Sc., B.Eng., A.O.S.M. and M.S. degrees from the Universities of New Zealand, Otago and Missouri. He immigrated to Canada in 1947 and for the next 17 years was employed in mining engineering and mine operations at Cominco Ltd., San Antonio Gold Mines Ltd., Hudson Bay Mining and Smelting Co., Limited and Upper Canada Mines Ltd.

In 1966, he joined the consulting engineering firm of Wright Engineers Limited and, as staff consultant, he was engaged in field investigations, feasibility studies and project design for mining projects in Canada, Mexico, Australia, Peru, Bolivia, Ireland, France, Morocco and Iran.

Mr. O'Hara returned to Hudson Bay Mining in 1970 and for the last nine years has managed the Development Group of senior engineers responsible for the evaluation of major mining and metallurgical projects, and corporate acquisitions for Hudson Bay Mining and Smelting and Anglo American Corporation of Canada.

Keywords: Orebodies, Evaluation, Open-pit mines, Underground mines, Process plants, Capital costs, Operating costs, Revenues, Equipment, Storage, Transportation, Housing, Manpower, Cost formulae.

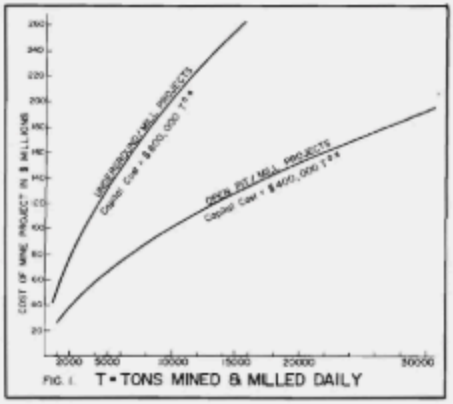


FIG. 1.

Estimates of the number of employees required, the operating costs per ton and some items of capital costs will require adjustment if the plant operates with seasonal shutdowns or if the mine and the mill operate on the same shifts per week.

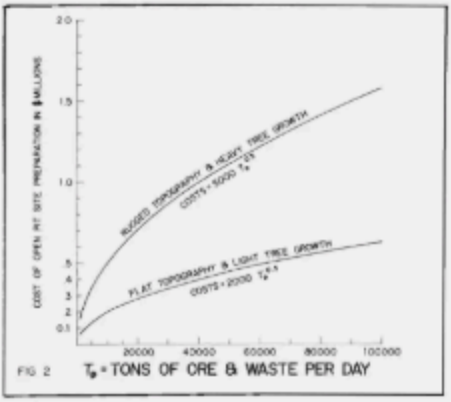


FIG. 2

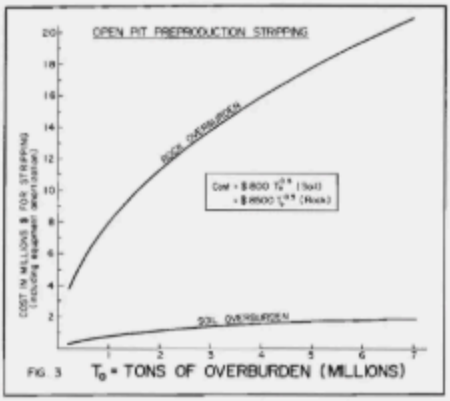


FIG. 3

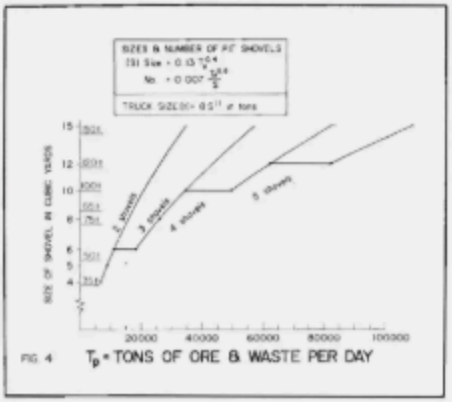


FIG. 4

Capital Cost Estimation—Open Pits

If the ore deposit is judged to be amenable to open-pit mining methods, an estimate must be made of the amount of soil and rock overburden that must be stripped before ore mining can be started, and an estimate must be made of the average ratio of waste tonnage to ore tonnage when the open pit is producing ore at the selected daily milling rate.

Site Preparation, Plant and Pit Roads

The cost of site preparation, as shown in Figure 2, will depend on the area, the topography and tree growth on the site of the proposed open pit, waste dumps and process plant. The area of the site and length of the access road to be prepared will typically vary with the square root of the size of the open-pit mine as expressed in terms of the tonnage of ore and waste (T_p) to be mined daily.

Pre-Production Stripping Costs

Figure 3 shows the cost of stripping soil and rock overburden, assuming that soil will be stripped by contractors using scrapers, and rock overburden will be removed by drilling, blasting, loading and haulage of rock by readily available sizes of drills, loaders and trucks. Because the size and condition of equipment readily available for rock overburden stripping is rarely suitable for production mining, the cost of this equipment must be amortized over the pre-production period.

Open-Pit Equipment—Sizing and Cost

The most important items of pit equipment are usually the pit shovels and haulage trucks, and because the size and number of haulage trucks depends on the size and number of shovels, selection of shovel size affects all other items of pit equipment. Figure 4 shows the typical selection of shovel size and number of shovels for different daily pit tonnages. Although some pits may have conditions suitable for partial replacement of shovels by front-end loaders at lower capital cost and improved flexibility, this study assumes that only shovels will be used for loading rock.

The size of haulage trucks should be matched against the shovel size so that a unit number of shovel loads fills the waiting truck. In general, the truck size in tons is about nine times the shovel size (in cubic yards) for small shovels, and about ten times the shovel size for large shovels as shown by the formula $t = 8\,S^{1.1}$.

The cost of open-pit equipment is shown in Figure 5 as a function of tons of ore and waste mined daily (T_P)

Figure 5 shows that the two major items in pit equipment cost are the fleet of shovels, each of which costs about

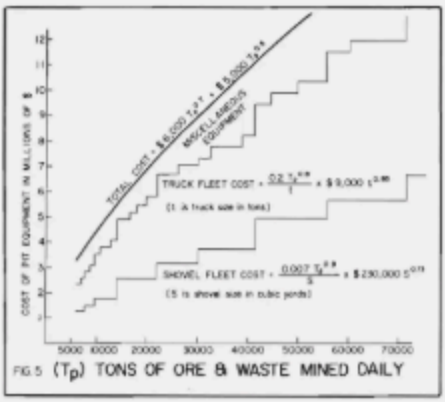


FIG. 5

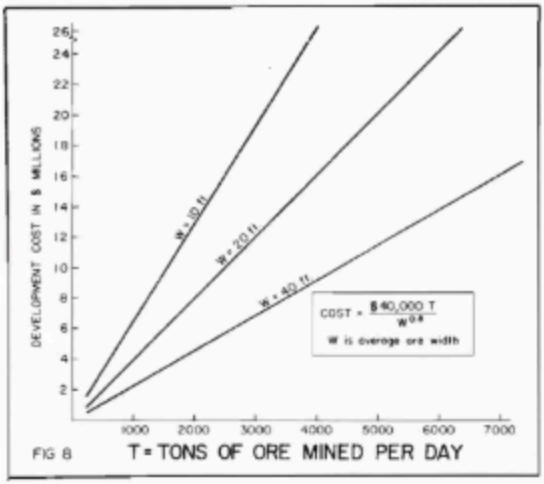


FIG 8

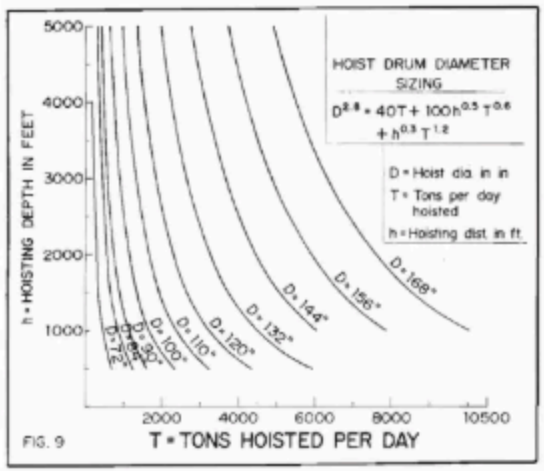


FIG. 9

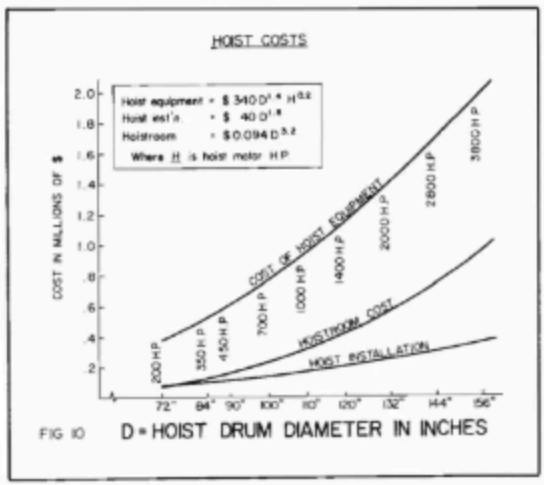


FIG. 10

on the use of one double-drum hoist for dual duty in hoisting ore and providing service facilities to all mine levels 24 hours per day. In practice, however, the larger mines, hoisting over 4000 tons of ore daily, will rarely depend on one large hoist for all skip hoisting and cage service requirements. Normally,

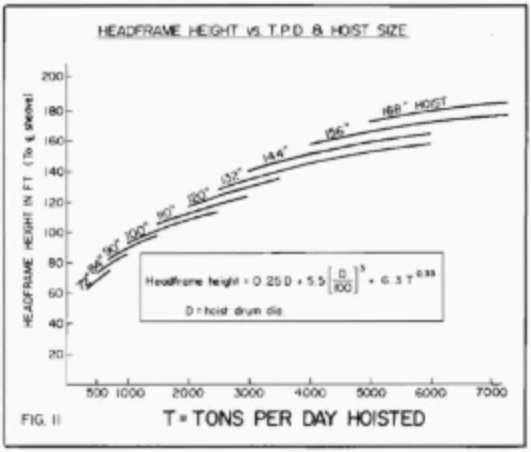


FIG. 11

large mines will use one hoist for ore hoisting only and a second hoist for cage service.

The drum size of a double-drum hoist required for hoisting T tons of ore daily from a hoisting depth of h feet is shown in Figure 9. The rated horsepower of the motor installed on the hoist will be dependent on the hoisting rope speed S (in feet per minute), but the formula shown in Figure 9 assumes that the appropriate hoisting speed for hoisting T tons of ore over a distance of h feet will be about $1.6\ h^{0.5}T^{0.4}$.

A hoisting speed slower (or faster) than this will require a hoist drum diameter slightly larger (or smaller) than D as computed by the formula in Figure 9, but the motor horsepower required would be somewhat smaller (or larger) than the size typically used on hoists of diameter D.

The motor horsepower suitable for driving a hoist of diameter D at a rope speed of S f.p.m. is:

Motor horsepower (H_1) = $0.5\ (D/100)^{2.4}S$

Figure 10 shows the capital cost of a fully equipped double-drum hoist of diameter D and H_1 motor horsepower, including all electric equipment. The cost of installing hoists, and the cost of constructing a hoistroom to house the hoisting equipment, is also shown in Figure 10. The hoistroom cost is based on the assumption that the hoistroom will have an area of $0.125\ D^{2.2}$ square feet, which will be sufficient for the installation of mine compressors in addition to hoisting equipment.

The height of the headframe must be sufficient to permit the skips to dump into orebins sized in relation to daily tonnage hoisted in addition to a safe distance for skip overtravel and hoist braking distance. Headframe height for different daily tonnages and hoist sizes is shown in Figure 11. The weight of structural steel (in pounds) in a headframe of height h feet and safely designed for hoisting ropes of 1/80 of the hoist drum diameter D will be approximately $0.12\ h^3\ (D/100)^2$. The cost of erecting the structural steel, headframe sheathing, piping and electrical equipment, deckhouse, sheaves, cages, skip dumps and orebins is shown in Figure 12.

Mine Compressor Plant

The size and cost of the mine compressors installed in the hoistroom building complex is shown in Figure 13. The size and consequently the cost of the compressor may be somewhat higher if the mine expects to utilize air-powered slushers and mucking machines instead of diesel-powered LHD equipment.

Underground Mining Equipment

The cost of equipment and equipment installation for underground development, stoping, loading, haulage, pumping, ventilation, crushing and miscellaneous mine services will increase as the daily tonnage increases, as shown in Figure 14. In the older conventionally equipped mines using portable

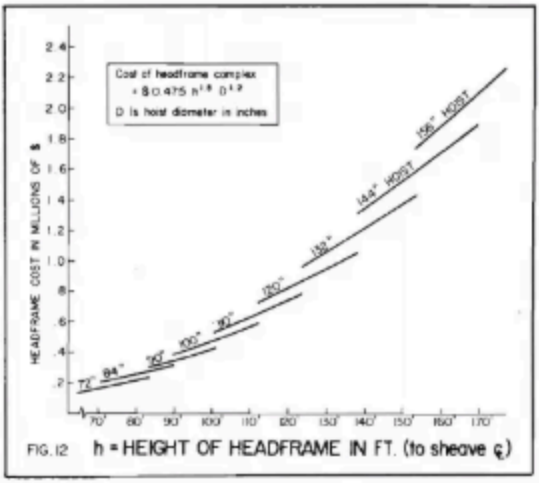


FIG. 12

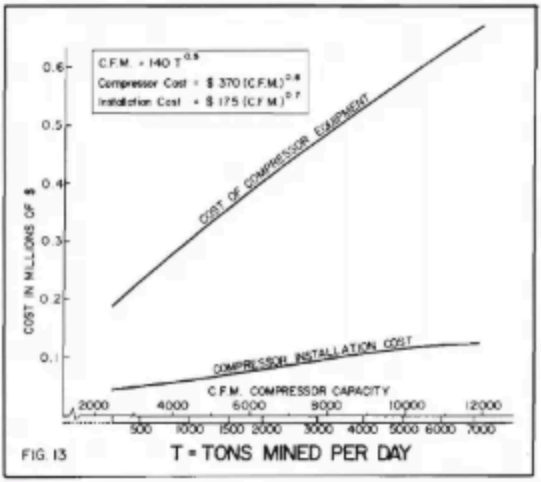


FIG. 13

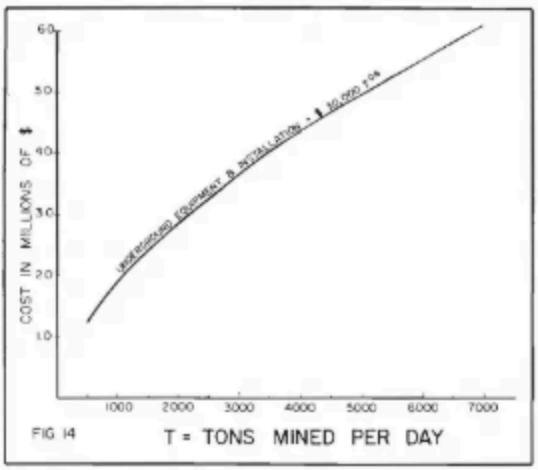


FIG. 14

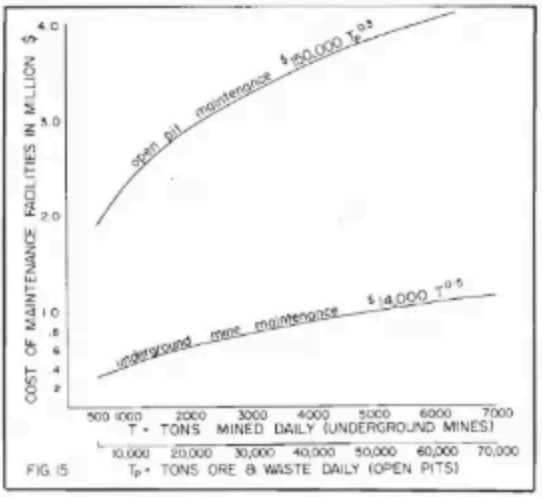


FIG. 15

jacklegs and stopers for drilling, slushers for stope mucking, mucking machines for drawpoint loading, and rail haulage, somewhat more equipment was required for mines with narrow stopes. If, however, modern mechanized equipment is employed whenever the stope width is adequate for its use, the cost of equipping a mine will depend primarily on the daily tonnage required.

Underground Mine Maintenance Facilities

The cost of constructing and equipping facilities for repair and maintenance of equipment drilling, loading, haulage and general mine services is shown in Figure 15.

It should be noted that the mine maintenance facilities would not include facilities required for maintenance of process plant equipment or general surface equipment. Most of the larger process plant equipment would be repaired within the mill repair bay, and only the smaller electrical items would be removed from the mill for repair in the general services maintenance facilities.

Capital Cost Estimation—Process Plants

Plant-Site Clearing and Mass Excavation

Clearing and excavation costs will vary widely depending on the topography and type of ground to be excavated, but for similar site conditions costs will vary with the plant-site area, which will be approximately proportional to the 0.3 power of the plant tonnage rate, as shown in Figure 16.

As compared with a flat plant site with less than 10 feet of soil overburden, excavation costs will be increased by 50% when the plant site is moderately sloping and some rock blasting is necessary to attain a level excavated area; excavation costs will be increased by 150% when the plant site is steeply sloping and extensive rock blasting is necessary to attain a terraced excavation area for the plant.

Concrete Foundations and Detailed Excavation

The cost of detailed excavation, compacted backfill, formwork, reinforcing steel, and concrete foundations for the plant and process equipment will be primarily dependent on the volume of concrete required. Mass excavation will have modified the localized site topography to levelled areas suitable for plant construction, and concrete volume does not depend on site topography.

The volume of concrete will, however, depend on the bearing capacity of the excavated ground; if this excavated ground is all in sound rock, concrete volume will be minimized, but if this excavated ground consists of compressible soil, the concrete volume and cost required will be substantially increased by the need for piled foundations and massive concrete slabs

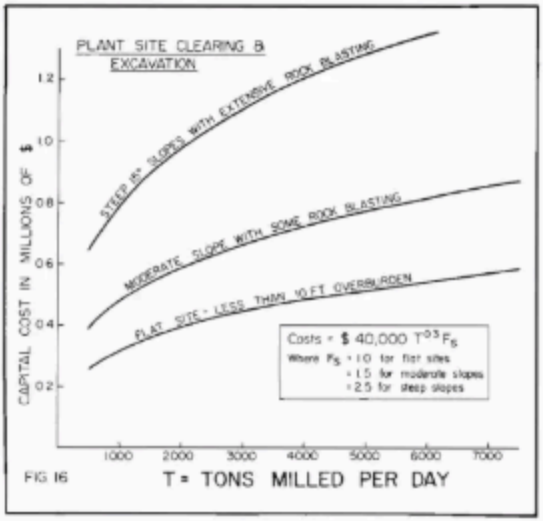


FIG. 16

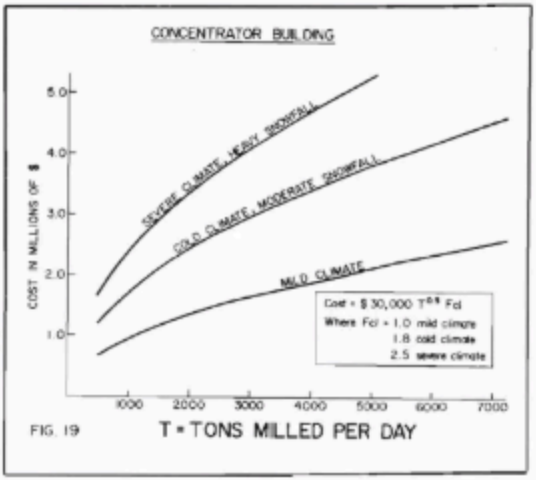


FIG. 19

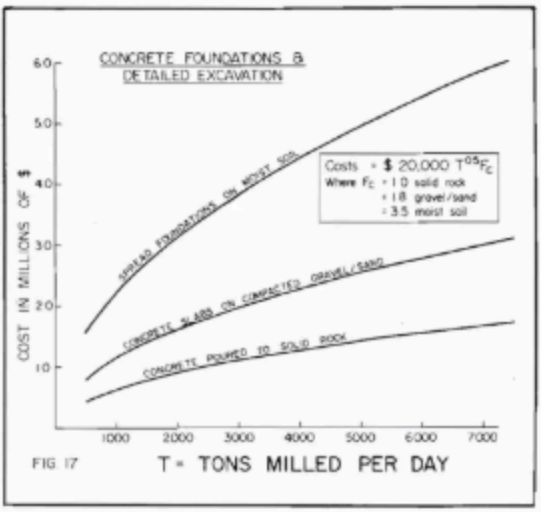


FIG. 17

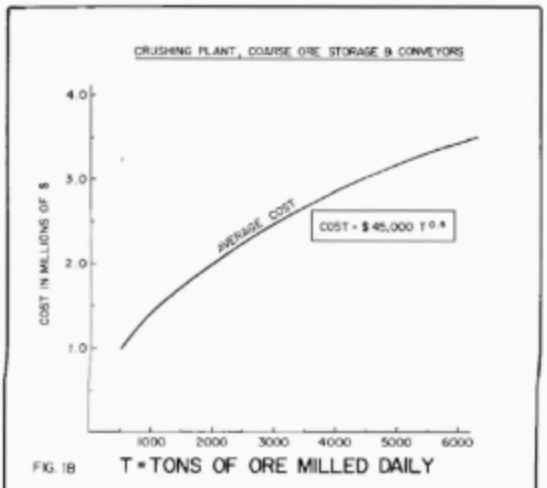


FIG. 18

to support the process equipment, as shown in Figure 17.

Crushing Plant, Coarse Ore Storage and Conveyors

For any specific tonnage rate, there could be many different design configurations of primary, secondary and tertiary crushing units linked by conveyors and ore storage to take advantage of site topography. In general, however, there will be an optimum design that minimizes capital cost in relation to daily tonnage capacity without compromising the availability of fine crushed ore for continuous 24-hour-per-day milling. This optimized capital cost will be approximately as shown in Figure 18.

Concentrator Building

Assuming that concrete foundations have been completed for the plant, the cost of constructing, sheathing, insulating and heating the concentrator building to house the fine ore bins, grinding section, flotation or cyanidation leaching section, thickening and filtration section, as well as offices, laboratories and supplies storage, will be approximately as shown in Figure 19.

The cost of the concentrator building will be substantially affected by the regional climate and winter snowload. In increasingly severe climates, building costs will be increased by the need for insulated sheathing and roofdeck, provision of heating equipment, structural strength to support snow loading or wind loading, and increased construction cost under severe climatic conditions.

Grinding Section and Fine Ore Storage

The capital cost of installing grinding mills, classifiers, fine ore bins and all accessory equipment for storage reclaim and grinding on previously prepared concrete foundations will be approximately as shown in Figure 20.

The cost of equipment and equipment installation will be increased when the ore needs fine grinding or is difficult to grind (i.e., has a high work index). Soft ores may be considered as ores which have a work index below 12, medium ores have a work index of about 15 and hard ores have a work index of 17 or greater. The fineness of grain of the ore and the degree of intergrowth with other sulphide minerals will determine the fineness of grind that will be necessary prior to flotation separation of the valuable minerals into several saleable concentrates. The fineness of grind is usually determined by metallurgical testwork on drill core samples, but a microscopic examination of core samples will normally suggest whether or not fine grinding will be necessary to separate several valuable minerals from the gangue.

It should be noted that the costs graphically shown in Figure 20 allow for sufficient fine ore storage to permit the mill to operate seven days per week on ore which has been crushed for five days per week. If the crushing plant will be operated each day that the mill is operating, less fine ore storage will be required and the cost of the fine ore storage and grinding section will be reduced by 25%.

Flotation and/or Processing Section

The capital cost of installing the processing equipment to concentrate the valuable minerals from the waste in the finely ground ore will depend on the type of separation process and the degree of complexity in the flowsheet. Figure 21 illustrates the cost of the process section for several different types of process and flowsheet complexity.

Thickening and Filtering Section

The capital cost of the thickening and filtration section depends on the volume of concentrates to be thickened and filtered, which in turn is a function of the process plant tonnage multiplied by the grade of ore being processed.

Capital cost = $\$5{,}000\ F_f T^{0.5}$, where the filtering factor:

F_f = 1.0 for low-grade straight Cu ores

F_f = 1.6 for high-grade Cu ores with recoverable zinc values

F_f = 2.0 for complex Pb-Zn-Ag ores or Cu-Zn-Pb ores

F_f = 3.0 for cyanided gold ores

Concentrate Storage and Loading

The capital cost of concentrate storage depends on the daily tonnage of concentrate T_c produced by the process plant. This can be computed by multiplying the daily ore tonnage processed by the mill recovery and by the ratio of the ore grade (or ore grades) to the concentrate grade (or concentrate grades).

Cost of concentrate storage and loadout = $\$4{,}000\ T_c^{0.8}$.

This cost would be appropriate for concentrate storage at the mill site from which concentrates will be transported to the smelter at frequent intervals by truck or rail. If, however, ocean shipment of concentrates will be envisaged, a tidewater storage and shiploading facility will be rquired. It may be possible to lease suitable facilities at an existing port; if leased facilities are not available, they must be constructed at a cost which may vary from several hundred thousand dollars for a small terminal at an existing port to several million dollars for a major terminal and port in a new location. It is not possible to suggest a rough cost for a tidewater concentrate terminal, because this cost will be greatly dependent on localized conditions of land access, tidal range, water depth, shipping access, etc.

Capital Cost Estimates for Plant Utilities and General Services

Electric Power Supply and Distribution

The peak load for mine/mill plants, as shown on Figure 22, increases with increased daily tonnage, but the rate of increase is slower for open-pit mines mining lower-grade ore than it is for underground mines which typically mine ore of higher grade requiring more complete processing.

The cost of electric power supply depends on the peak load requirements and also on whether the power is generated by a coal-fired plant (for large isolated mines) or by diesel generators (for small isolated mines), or is supplied by an existing utility through an extended transmission line and transformer station. The cost of each of these sources of electrical power is shown in Figure 22.

If utility power is available, an additional cost may be required to extend the transmission line to the plant site. This may cost about $60,000 per mile of line. For large mining projects consuming a large amount of electric power, the electric utility will typically install the transmission line and stepdown transformer station on the assured basis that future revenue from mine plant consumption of electric power will repay the

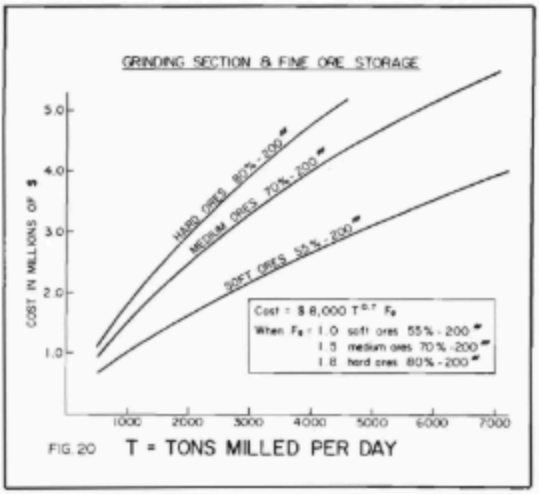


FIG. 20 T = TONS MILLED PER DAY

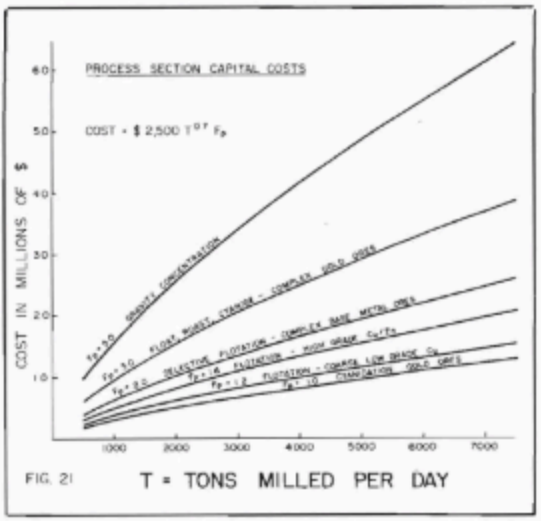


FIG. 21 T = TONS MILLED PER DAY

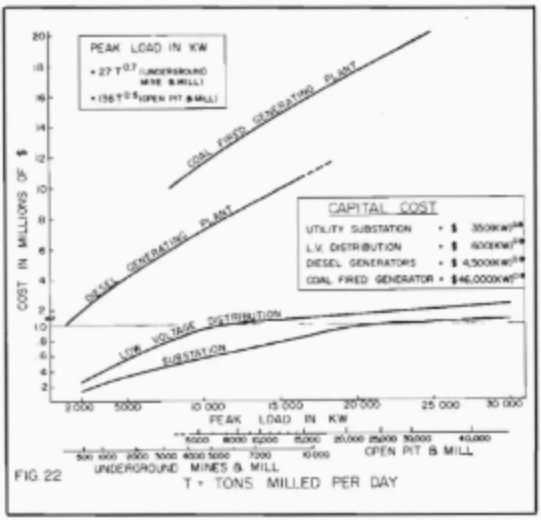


FIG. 22

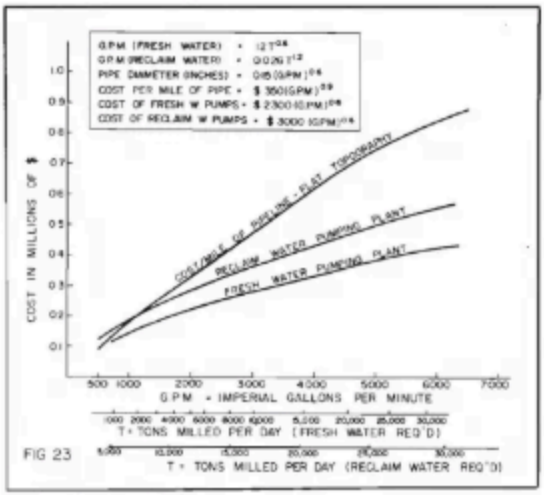


FIG 23

capital costs of the transmission line and transformer substation. For small mines, however, the mine may have to bear the cost of the extension of the transmission line and substation, either directly or effectively by payment of higher operating costs for electric power consumption during the initial years of operation.

The capital costs of the low-voltage electrical distribution from the main substation to the mine, mill and plant utilities will be approximately as shown in Figure 22. This cost will be common to all types of electric power sources: utility power, diesel or coal-fired generators.

Tailings Storage

The capital cost of pipelines and pumps to deliver mill tailings to the tailing storage area, and the cost of damming the storage area and preparation of settling and treatment ponds, is always difficult to estimate even when there is complete topographic data on the area selected for tailings storage. The cost of tailings storage can be drastically affected by environmental constraints and by stringent requirements imposed by regulatory agencies.

If it is judged that an environmentally acceptable tailings basin in moderate topography can be developed by damming with local material within 2 miles of the mill site, the cost of tailings storage for a mill treating T tons of ore daily will be about:

Cost = $8,000 $T^{0.5}$.

This cost represents the cost for which suitable tailings storage can be developed under favourable conditions of topography, accessibility and minimum environmental effects. Capital costs may, however, be many times larger when local topography is steep and rugged, when the selected tailings basin is many miles from the millsite and uphill pumping is required, or when the nature of the tailings imposes extremely stringent requirements for tailings treatment and containment.

The capital costs may be effectively increased by the administrative and technical costs of preparation of studies of environmental impact, planning of several alternative tailings storage facilities, and delay and uncertainty in obtaining approval of tailings storage planning.

Water Supply

Mines use large quantities of water for mining and milling and most of this water flows to the tailings storage, from which water reclamation may be necessary to avoid downstream pollution. The reclaimed water may be recycled in a portion of the mill circuit that is not adversely affected by the reagent content of the reclaimed water. The smaller mines frequently operate with fresh water only, whereas the larger mines, which could have much more effect on the regional stream drainage, are usually required to minimize their consumption of fresh water by recycling clarified water from the tailings pond.

In the Precambrian Shield area of Canada, where there is typically a plentiful supply of fresh water within a mile of the process plant, the volume of fresh water required for a mine which mills T tons of ore daily is about 12 $T^{0.6}$ gpm.

In the drier areas of British Columbia, where large-tonnage low-grade open-pit mines have been developed, water conservation is critically important and fresh water usage is typically only 2.5 $T^{0.6}$ gpm.

Figure 23 shows the typical requirements for fresh water and reclaim water for mines and mills of varying tonnages. The cost of the pump stations and pipeline cost per mile of pipe are also shown in the chart in Figure 23. These costs would be appropriate to the low-relief topography typical of the Precambrian Shield area of Canada, but in mountainous country, where high pumping heads are frequent, the cost of pump stations and pipelines would be much higher.

Capital Costs of General Plant Services

These costs include the costs of constructing and equipping the general administration offices, general warehouse, maintenance shops for the mill and surface facilities, vehicle garages, security stations and fencing, employee parking lots, changehouses and the cost of general-purpose vehicles. In general, these costs will vary as a function of the total number of company employees, which is not necessarily uniformly related to the plant capacity. The number of employees required for differing types of mining, milling and general services can be computed as described under operating costs, and, if this number is N employees, the total capital costs of general plant services can be computed as a function of N:

Costs of general plant services = $8,000 $N^{0.8}$.

Capital Cost of Access Road

If the mine is not readily accessible by an all-weather road, a road suitable for subsequent truck haulage of concentrates must be constructed before major project construction can get underway. A 30-ft-wide gravelled road with adequate drainage, curvature and gradients to permit satisfactory concentrate haulage conditions will cost roughly $200,000 per mile in regions with moderate topography, some rock outcrops and local sources of gravel. Under ideal conditions of flat topography, absence of exposed rock and well-drained gravelly soil, the cost per mile would be less; alternatively, in steep topography requiring heavy rock cuts, the cost of road construction could be much greater.

Bridges to span creeks or rivers where the total bridge length is L_b feet will cost approximately $130 $L_b^{1.5}$ (for each bridge).

Townsite and Housing Cost

If the mine site is within commuting distance of an existing town which has available housing and acceptable community facilities for N mine employees, the cost to the mining company may be limited to provision of housing for key staff only. This cost may be:

Existing townsite cost = $4,000 N.

Whenever a mining project is to be developed in an isolated region, a decision must be made as to whether the mine townsite will be developed as a bunkhouse camp (lower capital cost, high operating cost and transient work force) or as a developed townsite for family living (high capital cost, more attractive lifestyle, more permanent work force). Because a townsite must be of a certain minimum size before it can service the schooling, medical and recreational needs of families, the family townsite will be preferred only for mines employing in excess of 100.

When the number of employees (N) is less than 100, the bunkhouse camp will consist of single accommodation, mess

facilities and leisure facilities for 95% of the employees at $12,000 each as well as family accomodation for the remaining 5%, consisting of senior staff at $50,000 each, and minimum townsite facilities at 20% of housing cost:

Bunkhouse townsite cost = $20,000 N.

A family townsite accommodating 65% of the married employees in detached or apartment housing with facilities for schools, commercial, medical and recreational needs of the community as well as single accommodation for 35% of the employees and housing for townsite service employees will cost:

Family townsite cost = $55,000 N.

Project Overhead Costs

A. Feasibility studies, design engineering and technical planning: 4% to 6% of all mining pre-production costs, site preparation, excavation and road building costs, as well as 6% to 8% of all other project costs.

B. Project supervision, contract management, expediting and general construction facilities, including camp costs: 8% to 10% of all direct project costs.

C. Administration, accounting, legal and pre-production employment of key operating staff: 4% to 7% of all direct project costs.

D. Working capital for capital spares, supplies inventory and operating costs for the period between plant start-up and receipt of smelter payment for initial concentrate shipment; typically 4 months of operating costs on full production basis.

In general, the higher percentages for design engineering, project supervision and administration would be applicable to smaller mine projects, whereas the lower percentages would be applicable to major mine projects costing $100 million or more.

Estimation of Revenue

The revenue produced from mining and milling ore from an orebody that contains geologically estimated reserves of T_R tons grading G_R% metal is adversely affected by four factors:

1. Not all the ore reserve tonnage will be recoverable in practice by the planned type of open pit, or the practical shapes of underground stopes. It is difficult to offer any guide to the percentage of ore reserves that can be recovered by practical open-pit or underground stoping methods, because this percentage will vary greatly depending on the irregularity of the shape of the ore reserve blocks.

In general, if the ore reserve blocks are reasonably regular in shape and dip and are above the planned open-pit limits or the lowest underground level, a mineable recovery of over 95% of the ore reserve blocks could be expected.

2. Dilution of waste rock off the underground stope walls will vary with the type of stoping employed, the width of the stope in feet, the angle (A°) at which the stope is dipping and the competence of the stope wall rock. For stope wall rocks of average competence in relation to the type of stoping method employed, the dilution (D%) expressed as a percentage of waste rock in the mined ore will be approximately as shown in Figure 24.

When the stope wall rock is regular and competent, dilution may be only 0.7 of that shown in Figure 24, but if it is unusually weak and incompetent the dilution could be as much as 1.5 times the dilution shown.

Dilution has an adverse effect on operating costs because more tons must be mined to yield the same metal content as the undiluted ore, but it also has an adverse effect on revenue because the metal grade of each ton of diluted ore is reduced and mill recovery will consequently be somewhat lower than for undiluted ore.

Mill Recovery

The recovery of metallic minerals by flotation, cyanidation, leaching or gravity methods is affected by the fineness of grind needed to attain mineral unlocking, by the grade of the ore being milled and by the response of the mineral to the specific concentration process. Assuming that the valuable minerals can be fully separated from associated waste minerals at an economic fineness of grind, the recoveries and grades of metallic concentrates and precious metal ores will be about as shown by the formulae shown in Table 1. Somewhat higher recoveries will be attainable if the metallic ores are coarse grained, and lower recoveries will be inevitable if fine grinding does not fully unlock closely associated minerals, or if the valuable sulphide minerals are coated with oxidized minerals or slimes.

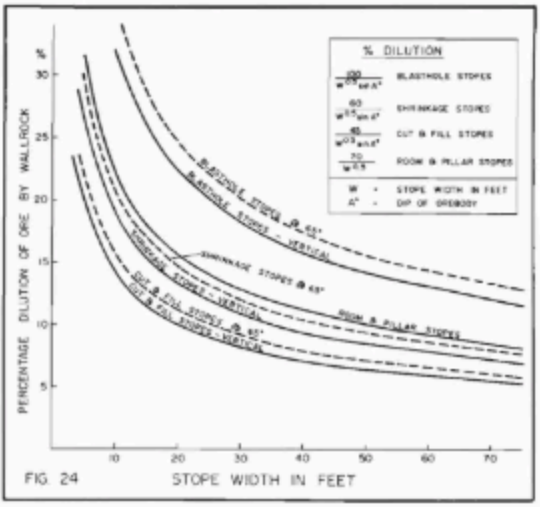


FIG. 24

Net Smelter Revenue for Base Metals (as of January 1979)

The net smelter revenue (at the mine), in $ per ton of base metal concentrates grading G% metal, is a function of the metal content of the concentrates (either a percentage P or a fixed unit deduction U) multiplied by a standard published metal price (M) less a specific deduction d in cents per lb (to allow for cost of refining and selling the smelted product), less a smelter charge S in $ per ton, less a freight cost F in $ per ton of concentrates transported from the mine to the smelter.

$ per ton = (P/100) $20 (G-U) (M-d)/100 - S - F

Typical Values:

Where P = 100% for Cu, 95% for Pb, 85% for Zn

U = 1.3 for Cu, 3 for Pb, 0 for Zn

M = 90¢ for Cu, 52¢ for Pb, 39¢ for Zn (as of January 1979)

d = 12.0¢ for Cu, 12.0¢ for Pb, 2.5¢ for Zn (as of January 1979)

S = $60 for Cu, $50 for Pb, $145 for Zn (approx.) (as of January 1979)

F = freight cost per ton of moist concentrates from mine to smelter, assuming that concentrates are trucked T_m miles by road, R_m miles by railroad and (if applicable) shipped by 15,000-ton freighter for D_o days of loading, ocean travel and unloading.

Approximate freight cost F in $ per ton of concentrates

$= \$0.17T_m^{0.9} + \$0.26\,R_M^{0.7} + \$0.80\,D_o$.

Estimation of Operating Costs

Operating Personnel—Open-Pit Mines

The number of men employed in open-pit operations, as shown in Figure 25, typically varies as the 0.5 power of the tons of ore and waste (T_p) mined daily, but the number of men required on truck haulage and road maintenance tends to vary with the 0.7 power of daily tonnage chiefly because, in the larger-tonnage pits, haulage distances are generally longer, and

TABLE 1. Base metal ores milled by flotation (head grades in %)

		Recovery Formulae	Typical Conc. Grade
Cu	in straight chalcopyrite ores	$R = 100\% (1\text{-}0.07\ Cu^{0.8})$	28.5% Cu
	in oxidized copper ores (sulphides)	$RCu_s = 100\% (1\text{-}0.08\ Cu_s^{0.8})$	Variable
	in oxidized copper ores (oxides)	$RCu_o = 100\% (1\text{-}0.40\ Cu_o^{0.3})$	Variable
	in copper-zinc ores	$R = 100\% (1\text{-}0.16\ Cu^{0.8})$	25.5% Cu
	in lead-zinc-copper ores	$R = 100\% (1\text{-}0.22\ Cu^{0.8})$	22.0% Cu
MoS_2	in straight molybdenite ores	$R = 100\% (1\text{-}0.04\ MoS_2^{0.8})$	88.0% MoS_2
	in copper-moly ores	$R = 100\% (1\text{-}0.06\ MoS_2^{0.8})$	Variable
Zn	in straight sphalerite ores	$R = 100\% (1\text{-}0.25\ Zn^{0.6})$	56% Zn
	in lead-zinc ores	$R = 100\% (1\text{-}0.32\ Zn^{0.6})$	53% Zn
	in copper-zinc ores	$R = 100\% (1\text{-}0.45\ Zn^{0.6})$	52% Zn
	in copper-lead-zinc ores	$R = 100\% (1\text{-}0.55\ Zn^{0.6})$	50% Zn
Pb	in straight galena ores	$R = 100\% (1\text{-}0.13\ Pb^{0.8})$	60% Pb
	in lead-zinc ores	$R = 100\% (1\text{-}0.18\ Pb^{0.8})$	53% Pb
	in copper-lead-zinc ores	$R = 100\% (1\text{-}0.28\ Pb^{0.8})$	45% Pb
Miscellaneous Ores			
Tungsten ores (gravity separation)		$R = 100\% (1\text{-}0.33\ WO_3^{0.5})$	75% WO_3
Nickel in nickel-copper ores		$R = 100\% (1\text{-}0.20\ Ni^{0.6})$	10% Ni
Uranium ores (flotation-leaching)		$R = 100\% (1\text{-}0.16\ U_3O_8^{0.8})$	77% U_3O_8
Iron ores (gravity-magnetic)		$R = 100\% (1\text{-}1.5\ Fe^{0.6})$	65% Fe
Precious Metal Ores and Precious Metals in Base Metal Ores: (head grades in oz per ton)			**Process**
Gold in siliceous ores		$R = 100\% (1\text{-}0.013\ Au^{0.8})$	Cyanidation only
in pyrite ores		$R = 100\% (1\text{-}0.03\ Au^{0.8})$	Flot./Roast./Cyan.
in base metal ores		$R = 100\% (1\text{-}0.3\ Au^{0.8})$	Flotation
Silver in straight silver ores		$R = 100\% (1\text{-}0.22\ Ag^{0.6})$	Flot./Grav. Separation
in base metal ores (-1.0 oz/t)		$R = 100\% (1\text{-}0.40\ Ag^{0.6})$	Flotation

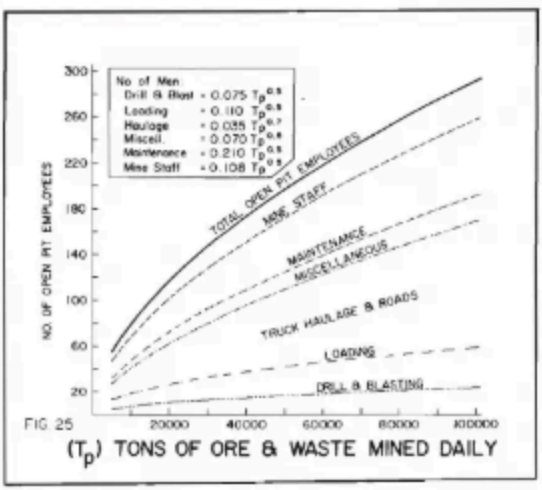


FIG. 25

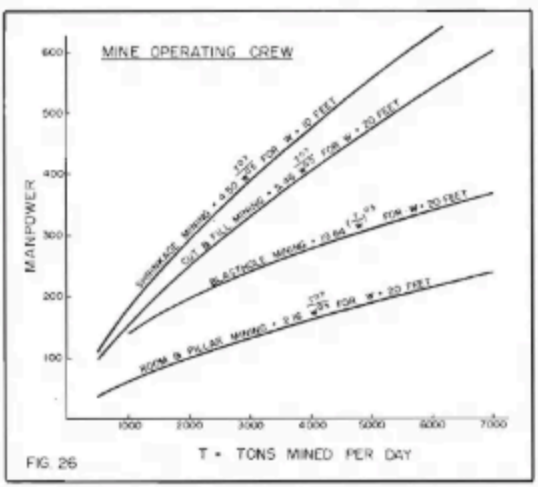


FIG. 26

consequently the ratio of men employed in haulage as compared with those employed in other pit operations tends to increase in larger open pits.

Operating Personnel—Underground Mines

The number of men employed underground, as shown in Figure 26, varies for different stoping methods, different stope widths (W in feet) and different daily tonnage rates (T in tons per day) (Table 2).

TABLE 2. Underground mining personnel

	Percentage Distribution of Mine Crew			
	Blasthole	Cut & Fill	Shrinkage	Room & Pillar
Development	17%	12%	9%	14%
Stoping	23%	31%	31%	45%
Mine Service	22%	20%	29%	12%
Maintenance	23%	23%	18%	13%
Mine Staff	15%	14%	13%	16%
	100%	100%	100%	100%

Operating Costs—Open-Pit Mines

The operating costs per ton for open-pit mines are shown in Figure 27. Labour costs are based on basic pay rates as follows: operators - $9.50/hr, maintenance - $8.90/hr, and staff - $10.40/hr, and an allowance of 35% of basic pay has been added to allow for fringe benefits. Cost of supplies per ton decreases less rapidly than labour costs as the daily mined tonnage increases.

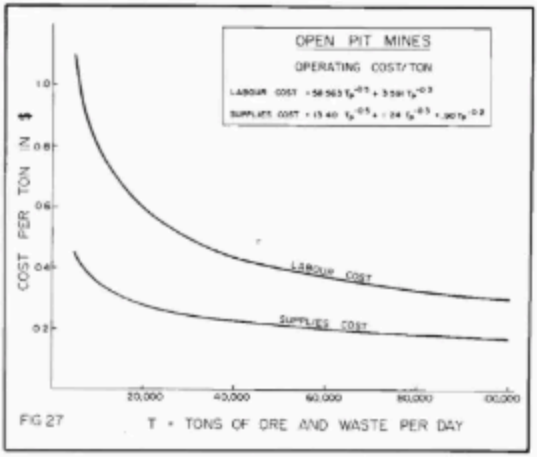


FIG 27

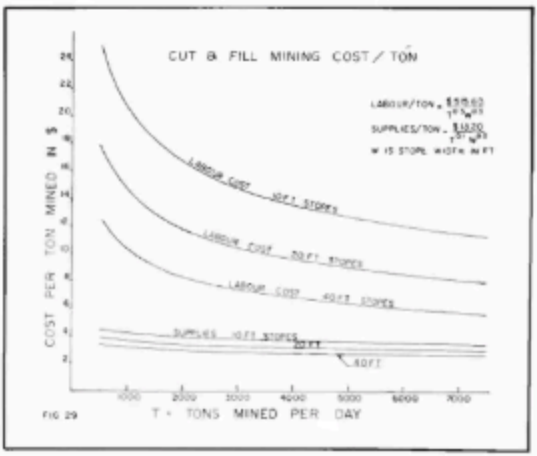


FIG 29

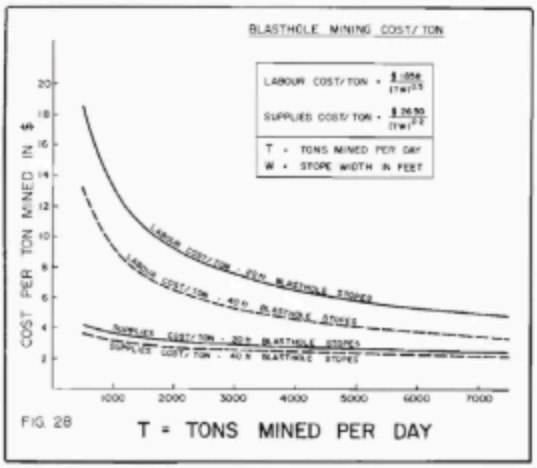


FIG. 28

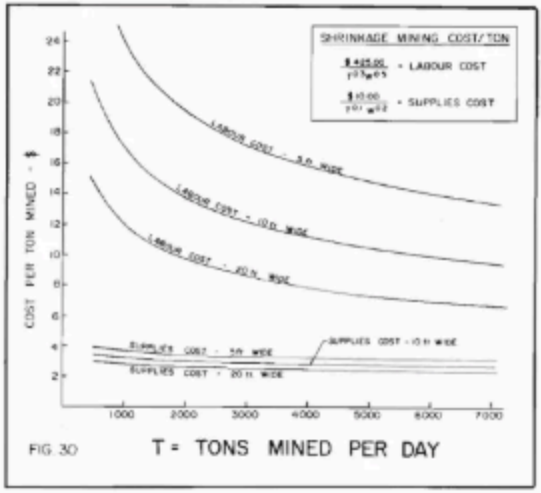


FIG 30

Operating Costs—Underground Mines

The operating costs for mines using blasthole, cut-and-fill, shrinkage or room-and-pillar stoping methods are shown on Figures 28 to 31. Labour costs are based on average basic pay rates of $8.50/hr for mine crew and $10.25/hr for mine staff, with an allowance of 35% to cover fringe benefits. The cost of supplies for development, stoping and general mine services typically varies with the number of men employed in these functions.

Employees Required—Process Plants

The number of mill employees and mill staff is shown in Figure 32 for differing process flowsheets. Typically, mills treating gold ores will require more employees in the cyanidation process than mills using flotation, and mills using selective flotation to produce two or more concentrates will require more employees in the flotation section than mills which produce only one flotation concentrate.

Mill Operating Cost per Ton

The operating costs of mills using differing processes are shown in Figure 33. Labour costs are based on an average pay of $7.40/hr for mill crew and $9.50/hr for mill staff, plus 35% to allow for fringe benefits. Grinding and crushing supplies normally constitute the largest item of mill supplies, and are also the most variable in terms of cost per ton because of differences in hardness and fineness of grind required for different ores.

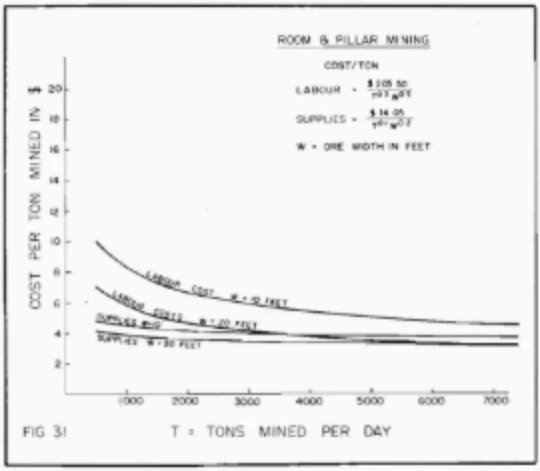


FIG 31

Administration and General Services—Employees

The employees required for administration and general plant services will tend to vary as a function of the total number of

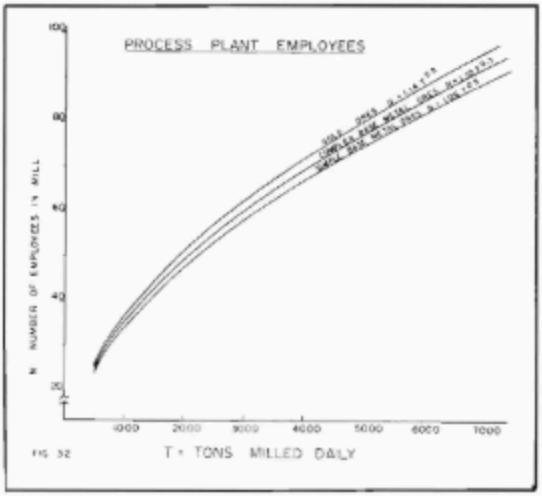


FIG 32

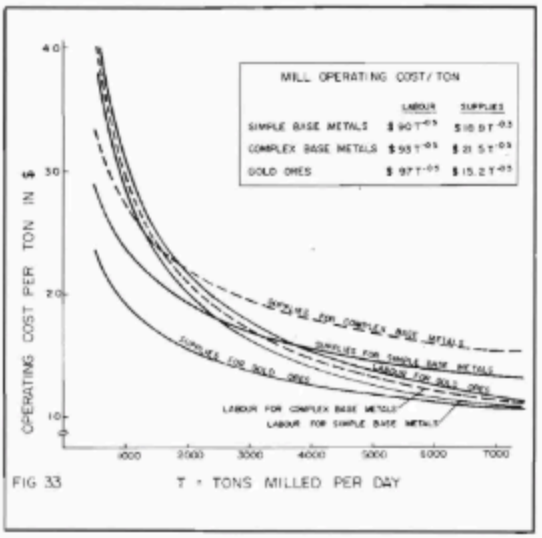


FIG 33

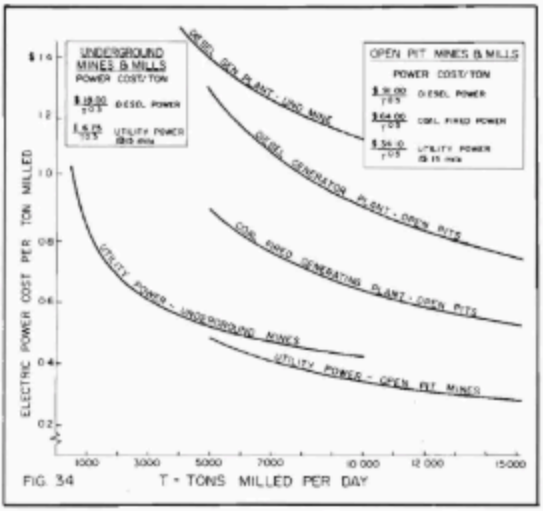


FIG. 34

staff, operating crew and maintenance crew in the mine and mill, and will be relatively independent of the tonnage throughput of the mine and mill (except insofar as the tons per day influences the size of the mine and mill crew).

Assuming total staff and crew of the mine and mill = Nm men

Employees required:

A. a) Electrical services = 0.03 Nm (power supplied by utility)
   b) Electrical services = 0.05 Nm (power diesel generated)
   c) Electrical services = 0.08 Nm (power generated by coal-fired plant)
B. Surface plant services & road maintenance = 0.04 Nm
C. (a) Townsite employees = none for existing town
   (b) Townsite employees = 0.03 Nm (bunkhouse-type mine camp)
   (c) Townsite employees = 0.05 Nm (family townsite—subsidized)
D. Concentrate transport (contracted out)
E. General administration = 0.07 Nm

Total employees N = Nm + (0.14 to 0.24)Nm, depending on type of townsite and electric power supply.

Administration & General Services—Operating Costs per Day

A. Electrical services—wages = $68 (0.03 to 0.08 Nm) (depending on type of electric power)
B. Surface plant services = $53 (0.04 Nm)
C. Townsite employees = $50 (up to 0.05 Nm) (depending on townsite type)
D. General administration = $85 (0.07 Nm)
E. Fringe benefits: 35% of above wages and salaries.
F. The electric power consumption and cost of electric power is shown in Figure 34 for underground mines and mills, and for open-pit mines and mills. Most of the electric power is usually consumed in the process plant, but the underground mine hoisting plant and compressor plant could be a significant consumer of electric power. Open-pit mines normally consume only a small fraction of the electric power required by the mill.
G. Supplies—general plant services = $6 $T^{0.5}$ per day
H. Townsite operating cost
   (a) = $13 Nm (bunkhouse camp subsidy)
   (b) = $ 5 Nm (subsidized family townsite)
I. General administration expenses, including office and warehouse supplies, telephone, travel expenses, property taxes, insurance, and legal, auditing and consulting fees:
   Cost per day = $4 Nm

Notes on Computation of Formulae

Virtually all the formulae shown in this paper for estimating size, quantity and cost of the major components of a mine project have been developed by the author from detailed data on many Canadian and foreign mining projects over the last fifteen years. Size and quantity data from foreign projects were used in determining requirements for mine projects under differing physical conditions of climate, topography, orebody shape, etc., but cost data from foreign projects were utilized only when unit costs were judged to be comparable to Canadian unit costs. Cost data from completed Canadian mine projects and foreign projects with comparable unit costs have been escalated to 1978 by using appropriate indices, and expressed in terms of 1978 Canadian dollars, which had an average 1978 exchange rate of $1.00 Canadian = $0.877 U.S.

Most of the operating cost data utilized in computing operating cost formulae were from Canadian underground mines in the Canadian Precambrian Shield, or from open-pit mines in Western U.S.A. and Canada. Operating cost data from foreign mines, where the cost of labour and supplies was very different from that of North America, were not used in computing average operating costs, but foreign data were utilized in analyzing trends relating consumption of supplies and labour to increases in tonnages mined and milled.

The relationships between mine project requirements and costs relative to plant capacity, operating costs of supplies and labour relative to daily tonnage mined and milled, and operating performance and mill recovery relative to mill-head grade, were determined by computerized statistical analyses of

the best fit of the data to an equation of the form $Q = KT^x$, where Q represents the actual data on quantities required or cost, and T represents the tonnage rate, milled head grade or other physical condition causing changes in quantities or costs.

The x values were determined to yield the lowest range of variation in K values across the widest range of T values for which reliable data were available. The x values tested were within a relatively narrow range which was judged to be consistent with technical considerations and operating experience. Whenever changes in the quantity Q were judged to result from simultaneous changes in two physical conditions T_1 and T_2, the K_1 values were first determined for the most influential condition T_1 to fit $Q = K_1T_1^x$, and then changes in the K_1 values were computed to fit the equation $K_1 = K_2T_2^y$, so that $Q = K_2T_2^yT_1^x$ where K_2 is a constant.

Because of the inevitable dispersion of data, the mathematical analyses did not rigorously consider each data point of equal value, but judgment factors were utilized in weighting data from individual mine projects according to the presence or absence of localized conditions that would result in unusually high or low values for costs or quantities.

International conference on energy and the environment

Envitec 80, which will be held at the Dusseldorf Fairgrounds (West Germany), **February 11-15, 1980,** is the world's largest and most comprehensive environmental technology exposition. During the fair, an international conference will be held on February 11, 12 and 13, 1980.

This meeting will bring together leading authorities from all over the world who will present the latest developments in environmental and energy conservation technology. It will cover a wide variety of subjects that have a direct impact on many of the world's common problems.

The speakers will represent government, industry, research and leading academic institutions from all over the world. The subjects cover energy and environmental policies, current state-of-the-art environmental processing technology, development of alternative energy sources and their impact on the environment, coal gasification, industrial pollution control (toxic gas, dust control, chemical waste removal and disposal), refuse disposal and recycling, refuse-fueled power plants, radioactive waste disposal, new energy-saving heating methods (new types of heat pumps, solar energy systems), alternate methods of power for both commercial and private vehicles (including new automobile engine design), air, water and noise pollution control, and other subjects that have a direct bearing on today's environmental needs.

This comprehensive conference deals with all aspects of the environment as related to international, community and industrial needs. It will provide new approaches and immediate solutions to many different individual problems. The assembly of the world's leading authorities on energy and the environment is of importance to anyone directly or indirectly involved with saving the quality of the environment. All presentations will be simultaneously translated into English as well as other languages.

The Envitec 80 International Trade Fair will attract over 30,000 visitors from over 50 different countries. The visitors will represent a high-level international audience made up of industry executives, government (local and national) officials, and research and academic institute representatives who will come to Dusseldorf to seek solutions to current and long-range environmental and energy problems. The fair will include over 400 different exhibits of equipment, processes and services covering every aspect of environment protection and control technology.

There will also be an Environment Info-Center during the full five days of Envitec 80, which will allow an exchange of information among industry, government and science. The Environment Info-Center will offer academic institutes, industry associations, government officials and other authorities in the field an opportunity to present the results of research as well as new operational technology. It will include open discussions with an exchange of experiences and viewpoints.

Exhibitors' seminars will also be held each day during Envitec 80. These seminars will allow the exhibiting companies to present and demonstrate new technology and procedures covering special subjects. As small-scale forums, the seminars will cover specific subjects such as air pollution control, water pollution control, noise abatement, solar energy, radioactive waste disposal, etc.

Workshop on the forming of aluminum sheet, plate and shaped products

A workshop on the forming of aluminum, sheet, plate and shaped products will be held **March 18-19, 1980** at the Los Angeles Hilton, Los Angeles, California.

This new workshop will be staged by the Society of Manufacturing Engineers and co-sponsored by the Aluminum Association and the National Association of Aluminum Distributors.

The emphasis of this two-day event will be to increase the productivity of aluminum users by providing practical guidance on the selection, handling and forming of aluminum mill products. The workshop will be divided into four sessions:

Session I. A technological base including industry nomenclature and terminology, alloys and tempers, material handling, finishing and segregation, and recovery for recycling.

Session II. Selection and application of lubricants for the forming of aluminum mill products will be detailed. Cutting of aluminum will cover the classification of cutting operations, blanking, tool design, force and clearance requirements, and equipment applications.

Session III. Roll forming technology and support equipment, including general applications, power specifications, tool design and equipment requirements, will be covered. Bending and forming of extruded shapes and tube products will treat bend limits of particular alloys.

Session IV. Press brake forming will include factors in choosing and using press brakes to form aluminum, their capabilities and limitations, and part, tool and die design. Drawing aluminum sheet will cover stages of drawing, characteristics and defects, blank development, redraw, complex drawing and secondary operations.

The fee for this two-day program is (U.S.)$195.00, covering all session materials, breaks and luncheons, and a special idea exchange reception planned for March 18, 1980.

For further information, contact David Johnson at the Society of Manufacturing Engineers, One SME Drive, Dearborn, Michigan 48128.